

demands of being able to quickly store and retrieve enormous quantities of data are ever increasing.

Scientists are now exploring hitherto uncharted territories, delving into advanced sciences such as holography, nanotechnology and advanced lasers. The implementation of these technologies will result in an increase in storage capacities, which is equivalent to what we've seen over the last few decades!

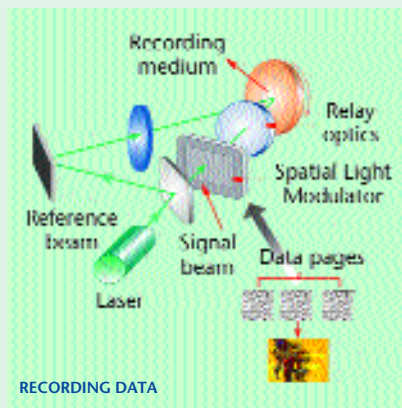
Research in these fields has created storage solutions such as holographic storage, blue laser and Millipede—each offering digital storage densities that are orders of magnitude greater than anything we know and use today. Even though devices that use these technologies are still on the outskirts of implementation, they are slowly and surely advancing down the road to commercial availability. The implications of what these technologies promise us is as dramatic as when we compare any of the commercially available gigabyte solutions today with the clunky sub-10 MB storage devices that were first used in computing!

Holographic storage

Consider the maximum data storage limits we have reached today: 15.9 GB on a double-layer, double-sided DVD? 180 GB on the largest hard disks? Think about a system that can store about a terabyte of information (that's over 1,500 CDs!) on a crystal the size of a sugar cube! Sounds like science fiction? Today, this is closer than ever to science fact! Holographic storage is a breakthrough technology that literally goes beneath the surface of the media and stores information within its volume. Current generation storage media store digital information on layers of platters either magnetically or optically. Holographic storage utilises the space within special types of crystals with optical characteristics, which are very similar to those used to create 3D holograms (the kind that you find in holographic stickers or on some product packages). Even though holo-

Inside a Holographic Data Storage System

In this simplified representation of a Holographic Data Storage System, the main component is a blue-green Argon laser,

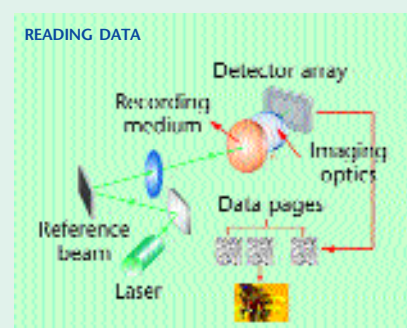


graphic storage has been around for over a decade now, giants in the field of technology such as IBM, Lucent and Rockwell are just beginning to make serious breakthroughs with this technology. Holographic storage incorporates the use of lasers, optics, image sensors and special crystals that respond to laser light.

Besides the astounding amount of data that can be stored using this technology, there are other inherent advantages that holographic storage offers. The

um-niobate crystal. To record the information, a second laser beam (the reference beam) is focussed on the exact same spot as the object beam that contains the data. When both these lasers meet inside the crystal, a phenomenon called interference occurs and an image of the data is created within the crystal in the form of a hologram.

To read back this page of information, the reference beam needs to be focused at exactly the same location as it was when the hologram was first created to store the data. The hologram of the data inside the crystal



diffraction this laser and the resulting page is projected onto a CCD (charge coupled device) sensor, much like the sensors used in digital cameras. The binary information can now be read from this CCD and the originally stored data is retrieved.

main advantage is the speed of retrieving the data—such systems will be able to retrieve data in tens of microseconds as compared to a data access time of almost 10 milliseconds offered by the fastest hard disks today. By the time they are commercially available, scientists say holographic systems could transfer an entire DVD movie in under 30 seconds! Another very important advantage is that of information search and retrieval. Consider the case of large databases that are

